

Module 1: Data Attributes & Proximity Metrics

KDD Pipeline, Attribute Types, Boxplots, Euclidean & Cosine Distance

Module I: Data Attributes & Proximity Metrics — Topics Covered

1. KDD Process & Data Mining Functionalities
2. Data Object Types & Attribute Categories
3. Five-Number Summary, Boxplots & Dispersion
4. Proximity Metrics: Euclidean, Manhattan, Minkowski
5. Cosine Similarity & Jaccard Coefficient

1. Proximity and Distance Metrics Formulations

Metric Name	Mathematical Formulation	Application Domain
Euclidean Distance (L_2)	$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$	Standard spatial coordinate distance for continuous numeric features.
Manhattan Distance (L_1)	$d(x, y) = \sum_{i=1}^n x_i - y_i $	Grid-like city block routing; robust against extreme outliers.
Cosine Similarity	$\text{sim}(x, y) = \frac{x \cdot y}{\ x\ \ y\ } = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2} \sqrt{\sum y_i^2}}$	Document text mining, TF-IDF vector comparisons (invariant to document length).
Jaccard Coefficient	$J(A, B) = \frac{ A \cap B }{ A \cup B } = \frac{q}{q + r + s}$	Asymmetric binary attributes (e.g., market basket item presence).